

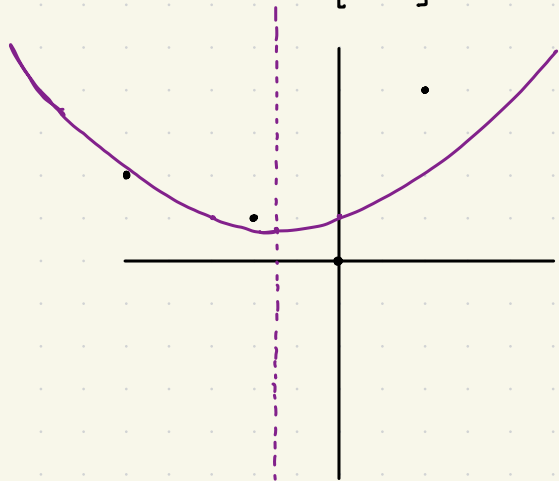
Assignment 12 Solutions

13.7 $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$

$$\begin{bmatrix} 1 & -5 & 25 \\ 1 & -2 & 4 \\ 1 & 0 & 0 \\ 1 & 2 & 4 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 4 \\ 6 \end{bmatrix}$$

$$X \vec{\beta} = \vec{y}$$

$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \vec{y} \approx \begin{bmatrix} 1.4 \\ 0.3 \\ 0.1 \end{bmatrix}$$



$$f(x) = 1.4 + 0.3x + 0.1x^2$$

13.10

$$\begin{bmatrix} \cos(-2) & \sin(-2) & -2 & 1 \\ \cos(-1) & \sin(-1) & -1 & 1 \\ 1 & 0 & 0 & 1 \\ \cos(1) & \sin(1) & 1 & 1 \\ \cos(2) & \sin(2) & 2 & 1 \\ \cos(3) & \sin(3) & 3 & 1 \end{bmatrix} \approx$$

$$\begin{bmatrix} -0.42 & -0.91 & -2 & 1 \\ 0.54 & -0.84 & -1 & 1 \\ 1 & 0 & 0 & 1 \\ 0.54 & 0.84 & 1 & 1 \\ -0.42 & 0.91 & 2 & 1 \\ -0.99 & 0.14 & 3 & 1 \end{bmatrix}$$

14.2

$$\begin{bmatrix} 3 & 3 & 1 \\ 3 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix}$$

14.3 $Q(x_1, x_2, x_3) = 2x_1^2 + 9x_2^2 + x_3^2 + 6x_1x_2$

14.4

$$\det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 9-\lambda & 3 \\ 0 & 3 & 1-\lambda \end{bmatrix} = (2-\lambda)((9-\lambda)(1-\lambda) - 9)$$

$$= (2-\lambda)(\lambda^2 - 10\lambda + 9) - 9$$

$$= (2-\lambda)\lambda(\lambda-10)$$

$$\lambda = 0, 2, 10$$

positive semidefinite

14.5

$$X X^T = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 3 & 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 1 \\ 0 & 1 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 15 \end{bmatrix}$$

$$\sigma_1 = \sqrt{15} \quad \sigma_2 = \sqrt{6}$$

14.6

$$A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = (\lambda - 17)^2 - 64$$

$$= \lambda^2 - 34\lambda + 289 - 64$$

$$= \lambda^2 - 34\lambda + 225$$

$$= (\lambda - 9)(\lambda - 25)$$

$$A^T A - 25I = \begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = x_2 \\ x_2 \text{ is free} \end{array}$$

$$\sigma_1 = 5 \quad \sigma_2 = 3$$

$$A^T A - 9I = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -x_2 \\ x_2 \text{ is free} \end{array}$$

$$\vec{v}_1 = \begin{bmatrix} r_2/2 \\ r_2/2 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} -r_2/2 \\ r_2/2 \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} \quad A \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -4 \end{bmatrix}$$

$$A = \begin{bmatrix} r_{1/2} & -r_{1/6} & -r_{1/3} \\ r_{1/2} & r_{1/6} & r_{1/3} \\ 0 & -2r_{1/2} & 1/3 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} r_{1/2} & r_{1/2} \\ -r_{1/2} & r_{1/2} \end{bmatrix}$$

$$\quad \quad \quad U \quad \quad \Sigma \quad \quad V^T$$

$$\vec{u}_1 = \begin{bmatrix} r_2/2 \\ r_2/2 \\ 0 \end{bmatrix} \quad \vec{u}_2 = \begin{bmatrix} -r_2/6 \\ r_2/6 \\ -2r_2/3 \end{bmatrix} \quad \vec{u}_3 = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 0 & -4 \end{bmatrix}$$

$$X^T \vec{u}_3 = \vec{0}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\begin{array}{l} x_1 = -2x_3 \\ x_2 = 2x_3 \\ x_3 \text{ is free} \end{array}$$

13.17 True unique LS $\Rightarrow$ LI columns $\Rightarrow$ A is invertible

13.18 False $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

13.19 True A is symmetric $\Rightarrow$ A is orthogonally diagonalizable
 $\Rightarrow$ A has an eigenbasis

$\Rightarrow$ all eigenspaces are as "large" as possible

13.20 False $A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

14.7 True 0 is not an eigenvalue

14.8 False $Q(x_1, x_2) = x_1 x_2 = \vec{x}^T \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vec{x}$

14.9 False $Q(x_1, x_2) = x_1^2 + x_2^2$ and $Q(\vec{x}) = 1$ for all $\vec{x}$

14.10 True $\|\vec{x}\|^2 = \vec{x}^T \vec{x} = \vec{x}^T \mathbf{I} \vec{x}$

14.14 $A - 4\mathbf{I} = \begin{bmatrix} -3 & 1 & 5 \\ 1 & 1 & 1 \\ 5 & 1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 4 & 8 \\ 0 & -4 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ $x_1 = x_3$
 $x_1 = -2x_3$
 x_3 is free

$A \vec{1} = 7 \vec{1}$ (by observation)

$\vec{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

(note that $\vec{v}_3$ is chosen to be orthogonal)

$A \vec{v}_3 = -4 \vec{v}_3$

$\min_{\|\vec{x}\|=1} Q(\vec{x}) = -4$ $\arg\min_{\|\vec{x}\|=1} Q(\vec{x}) = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$

$\max_{\|\vec{x}\|=1} Q(\vec{x}) = 7$

$\arg\max_{\|\vec{x}\|=1} Q(\vec{x}) = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$

$$14.16 \quad \textcircled{a} \quad \Sigma = \begin{bmatrix} 6\sqrt{2} & 0 \\ 0 & 3 \end{bmatrix} \quad V' = \begin{bmatrix} \sqrt{2}/2 & 0 \\ 0 & 1 \\ -\sqrt{2}/2 & 0 \end{bmatrix} \quad A^T = \begin{bmatrix} 2 & 2 & -2 \\ -4 & 2 & 4 \\ 4 & 1 & -4 \end{bmatrix}$$

$$\tilde{u}'_1 = \frac{A^T \tilde{v}'_1}{\sigma_1} \sim A^T \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \\ 8 \end{bmatrix} \quad \tilde{u}'_1 = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix} \quad A^T = U' \Sigma V'^T$$

$$\tilde{u}'_2 = \frac{A^T \tilde{v}'_2}{\sigma_2} \sim A^T \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \quad \tilde{u}'_2 = \begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix} \quad A = (U' \Sigma V'^T)^T = V' \Sigma U'^T$$

$$A = \begin{bmatrix} \sqrt{2}/2 & 0 \\ 0 & 1 \\ -\sqrt{2}/2 & 0 \end{bmatrix} \begin{bmatrix} 6\sqrt{2} & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}$$

$U \quad \quad \quad \Sigma \quad \quad \quad V^T$

$$\textcircled{b} \quad \|A^+ \tilde{b}\| = \|V \Sigma^{-1} U^T \tilde{b}\|$$

$$= \|\Sigma^{-1} U^T \tilde{b}\|$$

$$U^T \tilde{b} = \begin{bmatrix} \sqrt{2}/2 & 0 & -\sqrt{2}/2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 18 \\ -14 \end{bmatrix} = \begin{bmatrix} 12\sqrt{2} \\ 18 \end{bmatrix}$$

$$\Sigma^{-1} U^T \tilde{b} = \begin{bmatrix} \frac{1}{6\sqrt{2}} & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 12\sqrt{2} \\ 18 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$\|\Sigma^{-1} U^T \tilde{b}\| = \sqrt{40} = 2\sqrt{10}$$